



-1-

12. For cohesive soils, dewatering requires:
 A) Gravity drains
 B) Vacuum method or electro-osmosis
 C) Stone columns
 D) Blanket drains only
13. Which of the following is a purpose of dewatering during construction?
 A) Provide dry excavation
 B) Prevent heaving and piping
 C) Increase slope stability
 D) All of these
14. In a single-stage well point system, GWT can be lowered up to:
 A) 2 m
 B) 6 m
 C) 20 m
 D) 50 m
15. Electro-osmosis is effective in:
 A) Sandy soils
 B) Rocky strata
 C) Clayey and heterogeneous soils
 D) Lateritic soils only
16. Interceptor drains are used in:
 A) Airfield and highway pavements
 B) Mines only
 C) Forest land
 D) Agriculture only
17. The effectiveness of grouting depends on:
 A) Climate
 B) Soil fertility
 C) Type of roller used
 D) Groutability of soil/rock
18. Suspension grouts are:
 A) True solutions
 B) Mixtures of solids in water like cement grout
 C) Chemicals only
 D) Bitumen grouts
19. The basic steps in grouting are:
 A) Blasting
 B) Excavation, replacement, compaction
 C) Mixing soil with lime
 D) Drilling, cleaning, injection, monitoring
20. Which factor does not affect grout selection?
 A) Grain size
 B) Soil fertility
 C) Permeability
 D) Type of project

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) List any 8 factors to be considered while selecting a suitable ground improvement technique. Explain any two in detail.	8	L*3	1	1
	b) With neat sketches explain the vibro-flotation method of densification.	8	L2	2	1
2.	a) Define ground improvement. Explain the broad classification of ground improvement techniques with examples.	8	L2	1	1
	b) Define <i>compactive effort</i> and list factors affecting it in lab and field. List any three advantages and limitations of dynamic compaction.	8	L2	2	2
3.	a) Explain the objectives of ground improvement techniques. List different types of ground improvement techniques and explain any two in detail.	8	L2	1	2
	b) Explain the mechanism of densification during compaction. Briefly Elucidate the influence of compaction on permeability, compressibility, swelling, and strength.	8	L2	2	1
Unit – II					
4.	a) Explain the basic principle and concept of dewatering? Discuss its purposes during and after construction.	8	L2	3	1
	b) Briefly describe the following construction methods: (a) Mix-in-place, (b) Stationary plant, (c) Travelling plant.	8	L2	4	1
5.	a) With a neat sketch explain the steps in the design of filter material.	8	L2	3	1
	b) Explain the mechanism of soil-lime stabilization. Discuss the steps in construction of a lime-stabilized base course.	8	L2	4	2

6. a)	Describe hydraulic modification. List any four objectives of hydraulic modification. Also state the classification of drains.	8	L2	3	1
b)	Explain soil-cement stabilization? List out the applications and effects of soil-cement stabilization. Also explain any two factors affecting such stabilization technique.	8	L2	4	1
Unit – III					
7. a)	Explain the principle and effects of grouting. Discuss site considerations before starting grouting operations.	8	L2	5	1
b)	With relevant sketches explain the concept of soil nailing for the ground improvement.	8	L2	5	2
8. a)	With neat sketch, explain (i) permeation grouting (ii) compaction grouting (iii) Fractured Grouting.	8	L2	5	1
b)	Write a short note on different thermal methods of soil stabilization.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
